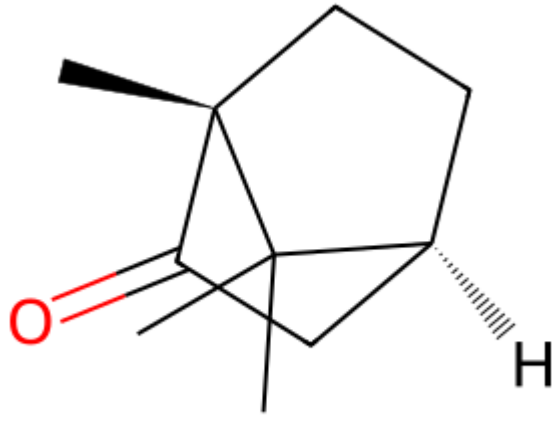


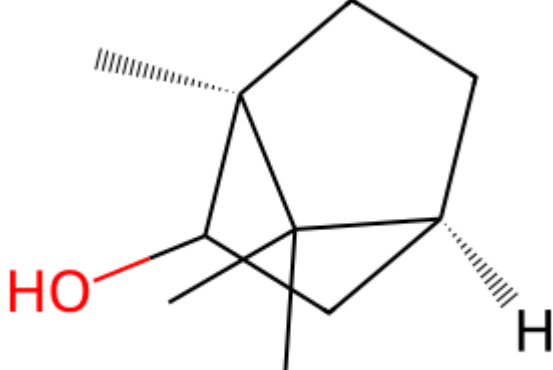


```
{'analysis_time': 0.0,  
'first_solution_time': 0.105919,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 4.737054346701574,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 4.737054346701574,  
'retro_star_value': 4.737054346701574}
```



Reaction

```
{'analysis_time': 0.105919,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 7.061601921848152,  
'retro_star_reaction_number': 7.061601921848152,  
'retro_star_rxn_cost': 0.021069475322770675,  
'retro_star_value': 7.061601921848152}
```

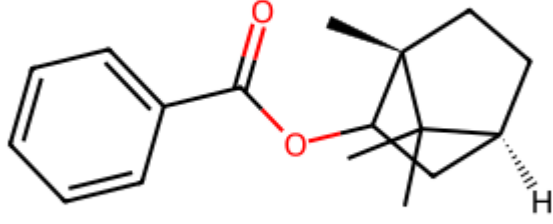


```
{'analysis_time': 0.105919,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 7.040532446525382,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 7.040532446525382,  
'retro_star_value': 7.061601921848152}
```



Reaction

```
{'analysis_time': 0.215599,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 22,  
'retro_star_min_cost': 7.040532446525382,  
'retro_star_reaction_number': 7.040532446525382,  
'retro_star_rxn_cost': 5.9488822546288755,  
'retro_star_value': 7.061601921848152}
```

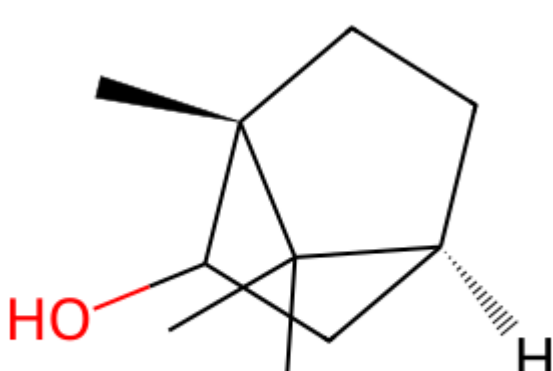


```
{'analysis_time': 0.215599,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 2,  
'num_calls_value_function': 22,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 1.0916501918965065,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 1.0916501918965065,  
'retro_star_value': 7.061601921848152}
```



Reaction

```
{'analysis_time': 1.231477,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 12,  
'num_calls_value_function': 316,  
'retro_star_min_cost': 1.0916501918965065,  
'retro_star_reaction_number': 1.0916501918965065,  
'retro_star_rxn_cost': 0.2808733523280516,  
'retro_star_value': 7.061601921848152}
```

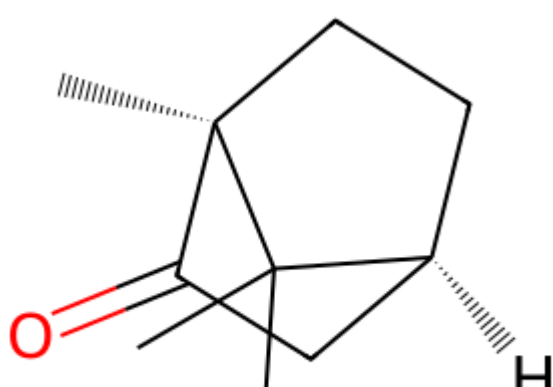


```
{'analysis_time': 1.231477,  
'first_solution_time': 1.483762,  
'num_calls_rxn_model': 12,  
'num_calls_value_function': 316,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 0.10229532290284653,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 0.10229532290284653,  
'retro_star_value': 7.061601921848152}
```

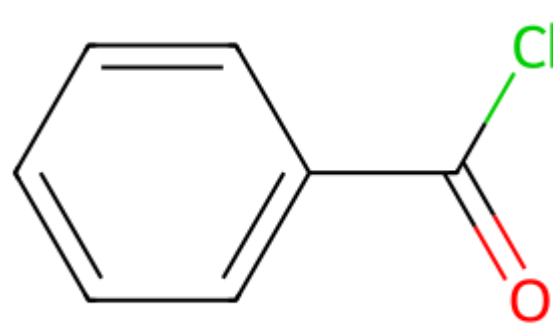


Reaction

```
{'analysis_time': 1.483762,  
'first_solution_time': 1.483762,  
'num_calls_rxn_model': 14,  
'num_calls_value_function': 371,  
'retro_star_min_cost': 0.10229532290284653,  
'retro_star_reaction_number': 0.10229532290284653,  
'retro_star_rxn_cost': 0.10229532290284653,  
'retro_star_value': 7.061601921848152}
```



```
{'analysis_time': 1.483762,  
'first_solution_time': 1.483762,  
'num_calls_rxn_model': 14,  
'num_calls_value_function': 371,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.061601921848152}
```

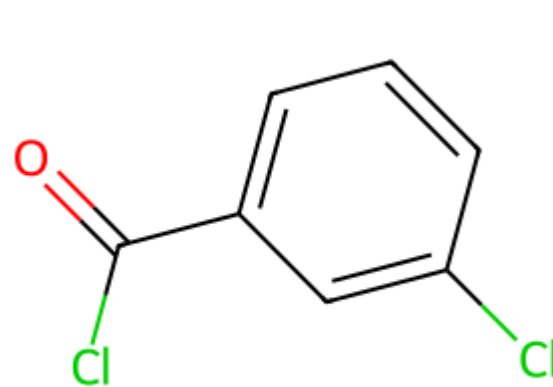


```
{'analysis_time': 1.231477,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 12,  
'num_calls_value_function': 316,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 0.7084815166656049,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 0.7084815166656049,  
'retro_star_value': 7.061601921848152}
```



Reaction

```
{'analysis_time': 1.534434,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 15,  
'num_calls_value_function': 412,  
'retro_star_min_cost': 0.7084815166656049,  
'retro_star_reaction_number': 0.7084815166656049,  
'retro_star_rxn_cost': 0.7084815166656049,  
'retro_star_value': 7.061601921848153}
```



```
{'analysis_time': 1.534434,  
'first_solution_time': 1.534434,  
'num_calls_rxn_model': 15,  
'num_calls_value_function': 412,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 7.061601921848153}
```